

# Christopher Musco

32 Vassar Street – Cambridge, MA 02139 – Office G578

☎ (401) 578 2541 • ✉ cpmusco@mit.edu

## Education

### Massachusetts Institute of Technology

Cambridge, MA

*Ph.D. Candidate, Computer Science – Theory of Computation*

2013 – present

Advisors: Jonathan Kelner, Martin Rinard

### Yale University

New Haven, CT

*B.S. Applied Mathematics, B.S. Computer Science*

2008 – 2012

AM Thesis: Fast Discrete Laplace Transforms (under Vladimir Rokhlin)

CS Thesis: Graph Constructions for Machine Learning (under Daniel Spielman)

## Research Interests

I am interested in dimensionality reduction and its application to randomized numerical linear algebra, sketching and streaming algorithms, and machine learning. I also work on problems related to algorithmic graph theory and collaborate with MIT's Center for Resilient Software on quantitative approaches to computer security and approximate computation.

## Academic Positions

### Massachusetts Institute of Technology

2013 – present

*Research Assistant*

## Teaching and Mentorship

### Research Science Institute/MIT

2014

*Summer Research Advisor*

Mentored Christopher Wang, a high school student participating in the prestigious RSI summer program.

- Designed a project on concurrent data structures under the direction of Professor Martin Rinard.
- Chris' paper, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

### Yale University

*Teaching Assistant, Introduction to Computer Science*

2012

Covered functional programming, boolean algebra, formal language theory, and basic complexity theory.

*Teaching Assistant, Introduction to Programming*

2011

Led weekly section for 15-25 students covering basic Java programming. Tutored 5-6 students individually.

*Undergraduate Grader, Introductory Calculus*

2009 – 2010

Scored problem sets and provided explanations and solutions for students, as well as feedback to the professor.

## Honors and Awards

### National Science Foundation

2014

*Graduate Research Fellowship*

### Yale University

2012

*Cum Laude, Distinction in Both Majors*

## Publications

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### Conference Papers.....

[Single Pass Spectral Sparsification in Dynamic Streams](#). Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *FOCS* 2014. (Invited to SICOMP Special Issue).

[Uniform Sampling for Matrix Approximation](#). Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *To appear in ITCS* 2015.

[Principled Sampling for Anomaly Detection](#). Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiropoulos-Douskos, Martin Rinard. *To appear in NDSS* 2015.

### Other.....

[Dimensionality Reduction for K-Means Clustering and Low Rank Approximation](#). Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. Preprint.

## Talks

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### **Single Pass Spectral Sparsification in Dynamic Streams**

Foundations of Computer Science (FOCS)

*Oct. 2014*

Harvard University, TOC Seminar

*Nov. 2014*

### **Dimensionality Reduction for K-Means Clustering**

Massachusetts Institute of Technology, Theory Lunch

*June 2014*

## Service

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### Reviewer.....

Network and Distributed System Security Symposium (NDSS)

*2015*

## Professional Experience

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### **Software Engineer, Data Team**

**Seattle, WA**

*Redfin*

*2012 – 2014*

Developed backend Java infrastructure for internet-powered real estate startup. Incorporated large-scale machine learning algorithms into production code base.

### **Elysium Digital**

**Cambridge, MA**

*Consultant*

*Summer 2011*

Provided technical expertise in software patent cases at leading litigation consulting company.

### **Developer**

**New Haven, CT**

*Amicus*

*2010 – 2011*

Built tools for political campaign management and social fundraising at early-stage startup.

### **Intern, Asset Finance Group**

**New York, NY**

*Moody's Investors Service*

*Summer 2009*

Established and monitored bond ratings on asset securitizations.

### **Intern, Natural Resources Conservation Service**

**East Greenwich, RI**

*U.S. Department of Agriculture*

*Summer 2007*

Collected and analyzed data for state conservation and research projects.